Standard Errors and Confidence Intervals for Total, Direct, and Indirect Effects in Continuous-Time Mediation Models

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Abstract

Mediation modeling using intensive longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive (DT-VAR) approaches are commonly used to estimate indirect effects in intensive longitudinal data (ILD), they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. To address these issues, continuous-time vector autoregressive (CT-VAR) models have been proposed as an alternative. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the total, direct, and indirect effects, for a range of time-intervals values, can be calculated using parameters estimated from CT-VAR models for causal inferential purposes. However, methods for calculating the uncertainty around the total, direct, and indirect effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results from using the delta and Monte Carlo methods to calculate standard errors and confidence intervals for the total, direct, and indirect effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed` `R` package.

Keywords: mediation, continuous-time, delta method, Monte Carlo method, R package, cTMed

Empirical Example

Data

In this example, we delve into an empirical data set employed by Ryan and Hamaker (2021). The dataset comprises four key variables: fatigue, self-doubt, irritability, and restlessness. These variables were originally extracted from a larger dataset used by Wichers et

al. (2016) and described in Kossakowski et al. (2017). Our primary objective is to investigate whether the association between a physical symptom (physical fatigue) and a cognitive symptom (self-doubt) is mediated by the affective symptoms (irritation and restlessness).

Before diving into the analysis, we followed a specific data preparation procedure, building upon Ryan and Hamaker's (2021) approach. Here are the steps we took. We constructed a new time vector by combining the smallest and largest time variables (0 and 5724.554, respectively). This new vector was incremented by the smallest empirical time interval (0.2255556). The newly created time vector was appended to the existing empirical time vector. We sorted the combined time vector and removed any duplicated values. For rows in the dataset lacking specific time values, we substituted missing values for the relevant variables. IS THERE A STANDARD WAY OF INSERTING MISSING VALUES? CITATIONS?

Estimated CT-VAR Drift Matrix

We employed the CT-VAR model using the `dynr` `R` package, aiming to closely align with Ryan and Hamaker's (2021) specifications. Despite using the same data set, we observed slight discrepancies in the drift matrix values compared to their study. These differences may arise from variations in estimation procedures and software packages. The estimated drift matrix is presented in Table 1.

The variables with the highest autoeffects are irritable and restless. The highest crosseffects is the path from restless to irritable. Crosseffects from irritable and restless to self-doubt are significant as well with a negative crosseffect for irritable and positive crosseffect for restless. The crosseffect from fatigue to restless is significant and negative.

PERHAPS ADD SOME THEORETICAL EXPLANATIONS FOR THESE ASSOCIATION AND HOW THESE RESULTS INFORMED THE MEDIATION MODEL TEST. TO BE HONEST, THE MEDIATION MODEL ON THE NEXT subsection WAS SELECTED BECAUSE IT WAS THE ONLY ONE WITH SIGNIFICANT RESULTS.

Mediation Analysis

Leveraging the estimated drift matrix and the corresponding sampling variance-covariance matrix, we calculated direct, indirect, and total effects of the model where the association between fatigue and self-doubt is mediated bi irritable and restless. We examined the mediated model across a range of time intervals (from 0 to 15). Confidence intervals were estimated using the delta and Monte Carlo methods. The results are presented in Figure 1.

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Table 1

Estimated CT-VAR Drift Matrix for the Empirical Example

	Self-Doubt	Fatigue	Irritable	Restless
Self-Doubt	$-1.1677(0.2162)^{***}$	$0.0474(0.0951)$	$-1.0385(0.4661)^{*}$	$1.2725(0.4797)^{**}$
Fatigue	$-0.0403(0.1451)$	$-0.7813(0.0734)^{***}$	$0.2177(0.3283)$	$-0.0185(0.3327)$
Irritable	$-0.6507(0.3203)^{*}$	$0.0867(0.1406)$	$-2.8225(0.7627)^{***}$	$2.0671(0.7578)^{**}$
Restless	$0.7989(0.2449)^{***}$	$-0.2632(0.1022)^{**}$	$1.4020(0.5106)^{**}$	$-2.0856(0.5352)^{***}$

Note. Point estimates with standard errors inside the parenthesis.
$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

Table 2

Estimated Direct, Indirect, and Total Effects for the Empirical Example at a Time-Interval of 1.7697

Effect	Point Estimate	Standard Error	z	p	2.5%	97.5%
			Delta Method			
Direct	0.0152	0.0299	0.5093	0.6106	-0.0434	0.0739
Indirect	-0.0488	0.0208	-2.3509	0.0187	-0.0895	-0.0081
Total	-0.0335	0.0286	-1.1718	0.2413	-0.0896	0.0226
			Monte Carlo Method			
Direct	0.0152	0.0315			-0.0509	0.0735
Indirect	-0.0488	0.0223			-0.0920	-0.0056
Total	-0.0335	0.0310			-0.0929	0.0281

Note. The indirect effect is significant for both the delta and Monte Carlo methods as the confidence intervals did not contain zero.

Notice that the results exhibit what is called an "inconsistent" mediation where the signs of the direct and indirect effects are opposite. Also, the total and direct effects are non-significant across the different time-interval values. For the indirect effect, there is a window of statistical significance where the confidence intervals did not contain zero. This window of significance centers around the time-interval associated with the absolute value of the indirect effect ($\Delta t = 1.7697$). Point estimates and confidence intervals for the point of maximal indirect effect is presented in Table 2.

PERHAPS ADD SOME THEORETICAL EXPLANATIONS FOR THESE RESULTS.

References

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, *23*(1), 61–75. https://doi.org/10.1080/10705511.2014.973960

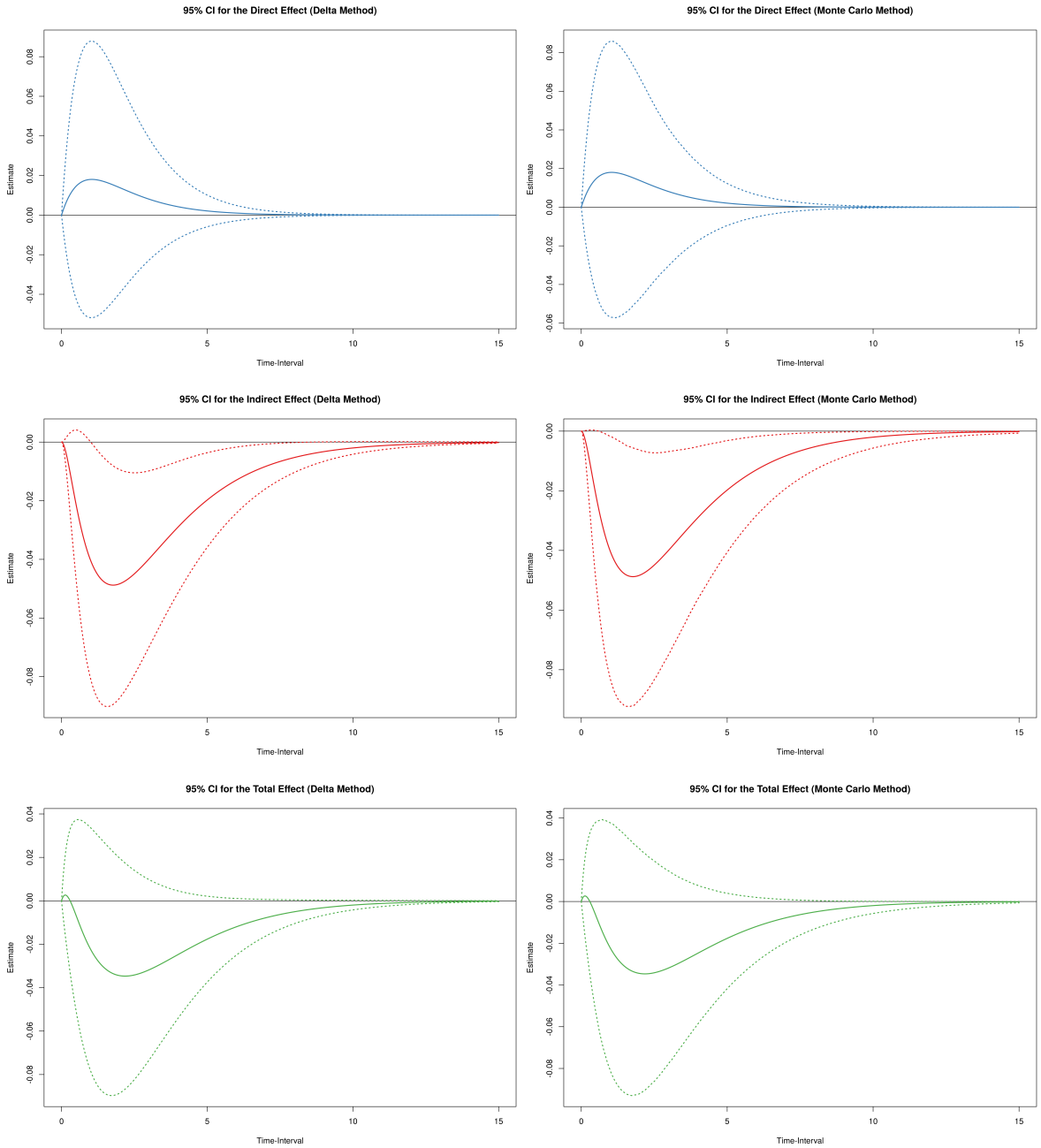

Figure 1

Estimates and 95% Delta Method and Monte Carlo Method (MC) Confidence Intervals (CI) of the Direct, Indirect, and Total Effects of Fatigue on Self-Doubt Through Irritability and Restlessness

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, *87*(1), 214–252. https://doi.org/10.1007/s11336-021-09767-0